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NeuroAgent

**A clinical collaboration proposal:
validating an evidence-driven neurological diagnosis benchmark**

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A new way of testing medical AI

Most research today tests medical AI with **question answering**. Real clinical work is not like that.

Today: question answering

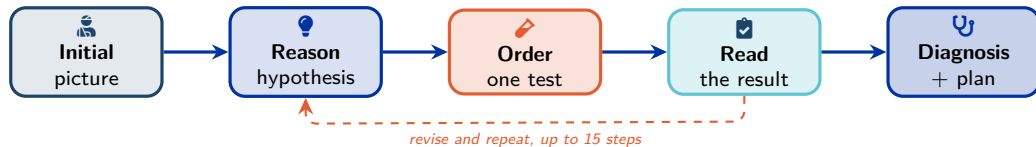
- Multiple-choice exams
- Every clue is already in the question
- One correct option to pick
- Rewards recall, not reasoning

Real clinical practice

- The patient arrives with a complaint, not options
- You decide which investigations to order
- Results come back over time
- Cost, risk and safety all matter

We introduce a **new paradigm**: test the AI on the **whole diagnostic workup**, the way a clinician actually reasons.

Project overview



600 realistic neurology cases across 20 conditions.

What we measure

- Right test, right order, sensible cost
- Sound reasoning, not memorised facts
- A workup a clinician can follow

Why it is new

- Today's AI exams hand over every clue
- Tests the real skill: what to investigate
- Each case checked by **2+ clinicians**

The agent in action

The goal: a transparent workup, ending in a diagnosis backed by evidence.

NeuroAgent (34383040)

Trace Replay / ALZ-EARLY-M01

68yo male ALZ-EARLY-M01
right-handed | Hispanic | BMI 27.3 | outpatient | 97% | 8/23

135/77 mmHg
Blood Pressure

Overview | History | Neuro Exam | Diagnostics | Ground Truth

CHIEF COMPLAINT
Subjective memory difficulties and word-finding trouble over the past year

HISTORY OF PRESENT ILLNESS
Mr. Eduardo Lopez is a 68-year-old right-handed retired civil engineer who presents to the neurology outpatient clinic accompanied by his wife, Carmen, for evaluation of subjective memory complaints that have been gradually worsening over the past twelve months. His wife first noticed the changes approximately a year ago when he began occasionally misplacing his reading glasses and wallet, which she initially attributed to normal forgetfulness. Over the ensuing months, however, she has observed that he sometimes struggles to find the right word during conversations, pausing mid-sentence and substituting a less specific word such as saying "the thing" instead of naming the object he intends. He has also become somewhat repetitive, asking the same question two or three times within an hour without apparent awareness that he has already asked. His wife notes that he has always been meticulous about the household finances, but over the past six months she discovered two unpaid utility bills and a duplicate payment to their cable company. When she brought this to his attention, he became frustrated and defensive, insisting he had taken care of everything. He continues to drive on familiar routes without incident, though his wife has noticed he seems more hesitant at complex intersections. He recently got momentarily confused in a large parking garage at the mall and had to use his phone to locate their car. He remains independent in all activities of daily living including bathing, dressing, and grooming. His appetite is unchanged and sleep is described as fair, though he uses his CPAP inconsistently. He has been less enthusiastic about his woodworking hobby and has mentioned feeling "foggy" some mornings. He denies visual hallucinations, personality changes, or aggressive behavior. There have been no episodes of getting lost while driving on well-known routes, no difficulty operating household appliances, and no falls. His wife denies witnessing any acting out of dreams or any limb jerking during sleep. He has not experienced headaches, seizures, tremor, or urinary incontinence. He has been somewhat more withdrawn socially in recent months but denies feeling depressed. He agreed to come to the appointment at his wife's insistence, acknowledging that "things aren't as sharp as they used to be."

Agent Timeline

- 1. **Agent Reasoning**
"Initial": This is a 68-year-old male presenting with progressive memory difficulties, word-finding...
- 2. **Lab Results**
clinical_context: 68-year-old male with non-onset cognitive decline; comprehensive dementia... [Done]
- 3. **Agent Reasoning**
"Initial": The lab results are informative but not surprising. Reversible causes have been ruled out...
- 4. **Brain MRI**
clinical_context: 68-year-old male with progressive memory impairment, word-finding difficulty... [Done]
- 5. **Agent Reasoning**
"The lab and MRI results now provide a comprehensive picture. Key findings include: [1] ..."
- 6. **Drug Interactions**
drug: donepezil, current_medications: ["Metformin", "Amlodipine", "Rosuvastatin", "Tamsulosin"]... [Done]
- 7. **Final Assessment**

Primary Diagnosis
Mixed Dementia (Alzheimer Disease + Vascular Cognitive Impairment) (Confidence: 0.80)

Differential Diagnoses

- Pure Alzheimer Disease** - Less likely due to significant vascular burden (Fazlous 2 white matter diseases, lacunar infarct, microbleeds) contributing meaningfully to cognitive profile.
- Vascular Dementia Alone** - Unlikely because hippocampal atrophy (MTA score 1-2) and temporal/parietal cortical volume loss are characteristic of AD pathology; vascular injury alone doesn't explain prominent episodic memory deficit.
- Psychogenic/Pseudodementia** - Ruled out by gradual progressive course over 12 months, objective neuropsychological testing showing true cognitive impairment (MMSE 26/30), preserved awareness of deficits.

Key Evidence

- Clinical Features**: Progressive episodic memory loss over 12 months; executive dysfunction (financial mismanagement, driving hesitation); word-finding difficulties—consistent with AD phenotype.
- Neuropsychological Testing**: MMSE 26/30 with specific deficits in delayed recall (epi-toxic memory), clock drawing (executive/aparal), borderline category fluency—pattern typical of Alzheimer's.
- MRI Findings (Crucial)**: Bilateral hippocampal atrophy (Scheltens MTA 1-2) AND moderate small vessel ischemic disease (Fazlous 2); chronic lacunar infarct in left caudate, scattered microbleeds—demonstrates concurrent AD and vascular.

The dataset

Cases

600

20 conditions

Per condition

30

exactly, fully balanced

Follow-up results

3,800

triggered by actions

How each case is built

- Patient core: history, exam, vitals
- Evidence hidden behind tool calls
- Ground truth: best path, red herrings, unsafe actions

Spanning

- 3 difficulty tiers: straightforward to puzzle
- 396 synthetic + 204 real-report-seeded
- 3 settings: emergency / inpatient / outpatient

*Results are cached in two tiers: **initial** findings when a test is first ordered, and **follow-ups** unlocked by a specific next action.*

How the cases were generated

Two pipelines, one schema: real cases for fidelity, synthetic cases for scale.

Seeded from real case reports

- Source: **MedCaseReasoning**, clinician-written case reports
- Rewritten by **Claude Opus 4.8 + Sonnet 4.6**: findings hidden in tool outputs

204 cases

Fully synthetic, for scale

- Built from per-condition clinical schemas, no source report
- Scale needed to **fine-tune local models**
- Same schema and ground truth as the seeded cases

396 cases

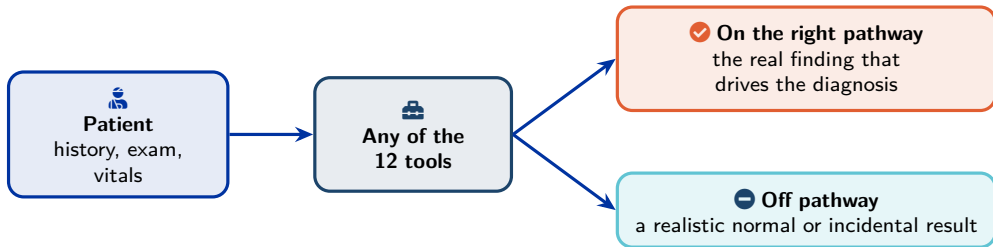
The 12 diagnostic tools

Tool	What it returns
Brain MRI	report with region-tagged findings
EEG	background, epileptiform features, focality
ECG	rhythm, intervals, signs of ischaemia
Labs	panels with reference ranges + abnormality flags
CSF (lumbar puncture)	cells, protein, glucose, NfL, oligoclonal bands
CT scan	non-contrast / contrast, head / spine
Echocardiogram	chambers, valves, ejection fraction, thrombus
Cardiac monitoring	telemetry / Holter / loop recorder
Advanced imaging	MRA / MRV / CTA / PET / DaT-SPECT
Specialised tests	EMG/NCS, biopsy, autoimmune panels
Literature search	guideline and paper snippets
Drug interactions	interaction profile + contraindications

Each returns a structured, typed report with a cost; any tool, any time.

Every patient is a complete picture

Each case pre-computes a **whole patient**, so the agent can order **any** of the 12 tools and always get a realistic answer.



Never a dead end: every tool answers. But each call costs money and a turn, so ordering the wrong test is **penalised**.

A worked example: wake-up ischemic stroke

Turn 1: what the agent sees

- 58 F: woke with slurred speech, clumsy hand
- Last known well 9 h ago: window unclear
- Left face/arm weak, sensory loss, neglect

What the right workup reveals

- CT first: no bleed; glucose normal → tPA cleared
- MRI: fresh infarct, deep right white matter
- ECG, echo, carotids, Holter: hunt the cause

The trap, and an off-pathway test

“Chronic, not acute” tempts an early stop, yet the cause hunt is still owed. An off-pathway EEG reads **normal**: wasted.

📌 **Ground truth:** acute ischemic stroke, right corona radiata (ICD-10 I63.541).

How we evaluate performance

Automatic metrics, per case

- **Diagnosis:** ICD top-1 / top-3 match
- **Tool selection:** precision, recall, F1 + coverage
- **Safety:** contraindicated, harmful, wrong-order calls
- **Cost:** \$ vs optimal; useless or repeated calls

AI reasoning judge, 0–5 each

- Diagnostic accuracy; evidence found + integrated
- Differential reasoning; uncertainty calibration
- Tool efficiency, safety, red-herring handling
- One **composite reasoning score**

Clinician validation: the gold standard

≥2 clinicians per case: field-level approval + inter-rater agreement, the human ceiling we calibrate against.

What we are asking of you

1. Validate the cases

- Read each patient as you would in clinic
- Approve, flag, or correct field by field
- Drop cases that are not good enough

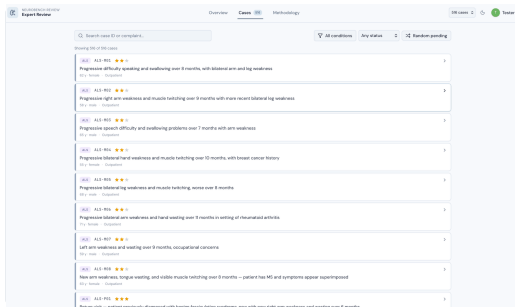
2. Validate our scoring

- Score a sample of agent runs yourself
- Compare against the AI reasoning judge
- Confirm scores track clinical quality

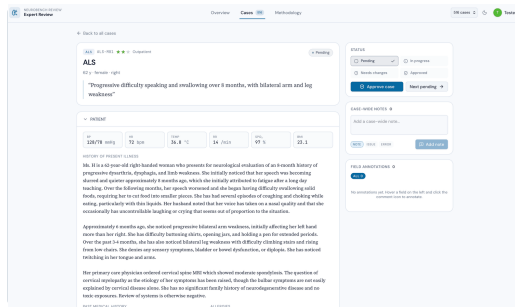
How the review works

- **≥ 2 independent reviewers** per case
- Separate login per reviewer: blind, no shared bias
- Agreement stats $\rightarrow$ fix or drop weak cases $\rightarrow$ higher quality

The annotation platform



 Browse cases: condition, difficulty, status



 One screen per patient: vitals, history, full workup

Reviewing a case

PAST MEDICAL HISTORY

Hypertension, diagnosed 10 years ago, well-controlled on lisinopril

Hypothyroidism, on levothyroxine for 8 years

Osteoporosis, on calcium and vitamin D supplementation

PATIENT > CLINICAL HISTORY
> PAST MEDICAL HISTORY > [2]

Note Issue Error

Describe the issue or note...

⌘+ to save - Esc to dismiss Cancel Add

Alcohol Social 1-2 glasses of wine per week

Living Lives with husband, two adult children

Situation nearby

Functional Previously fully independent; now difficulty

ALLERGIES

Penicillin — rash

MEDICATIONS

Lisinopril · 10 mg · once daily
for Hypertension

Levothyroxine · 75 mcg · once daily in the morning
for Hypothyroidism

Calcium carbonate · 600 mg · twice daily with meals
for Osteoporosis prevention

Cholecalciferol · 1000 IU · once daily
for Vitamin D supplementation

STATUS

☐ Pending ☒ In progress ✓

☐ Needs changes ☐ Approved

Approve case Next pending →

CASE-WIDE NOTES 0

Add a case-wide note...

NOTE ISSUE ERROR Add note

FIELD ANNOTATIONS 1

ALL 1 ISSUE 1

patient.history_present_illness ISSUE

age not coherent with the pathology

🗨 Flag any field: note, issue, or error

☰ Per-case status, every annotation
logged

Approve, request changes, or move on: progress tracked per reviewer.



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